A. R. Onojeghuo

PhD, Geography (Science)

Data Science Specialist|GIS & Remote Sensing Specialist | Geospatial Analysis Consultant

About me

Results-oriented Data Science, GIS & AI Specialist leveraging advanced geospatial analytics, machine learning, and cloud solutions. Experienced in transforming complex multi-source data into actionable spatial intelligence and clear technical insights.

GeoAI & Data Science

- **Python & R:** GeoPandas, PyTorch, ggplot2, tidyverse
- **Spatial ML:** Predictive modeling, land cover classification, Spatial AI
- **Database & Cloud:** SQL, BigQuery GIS, spatial query optimization

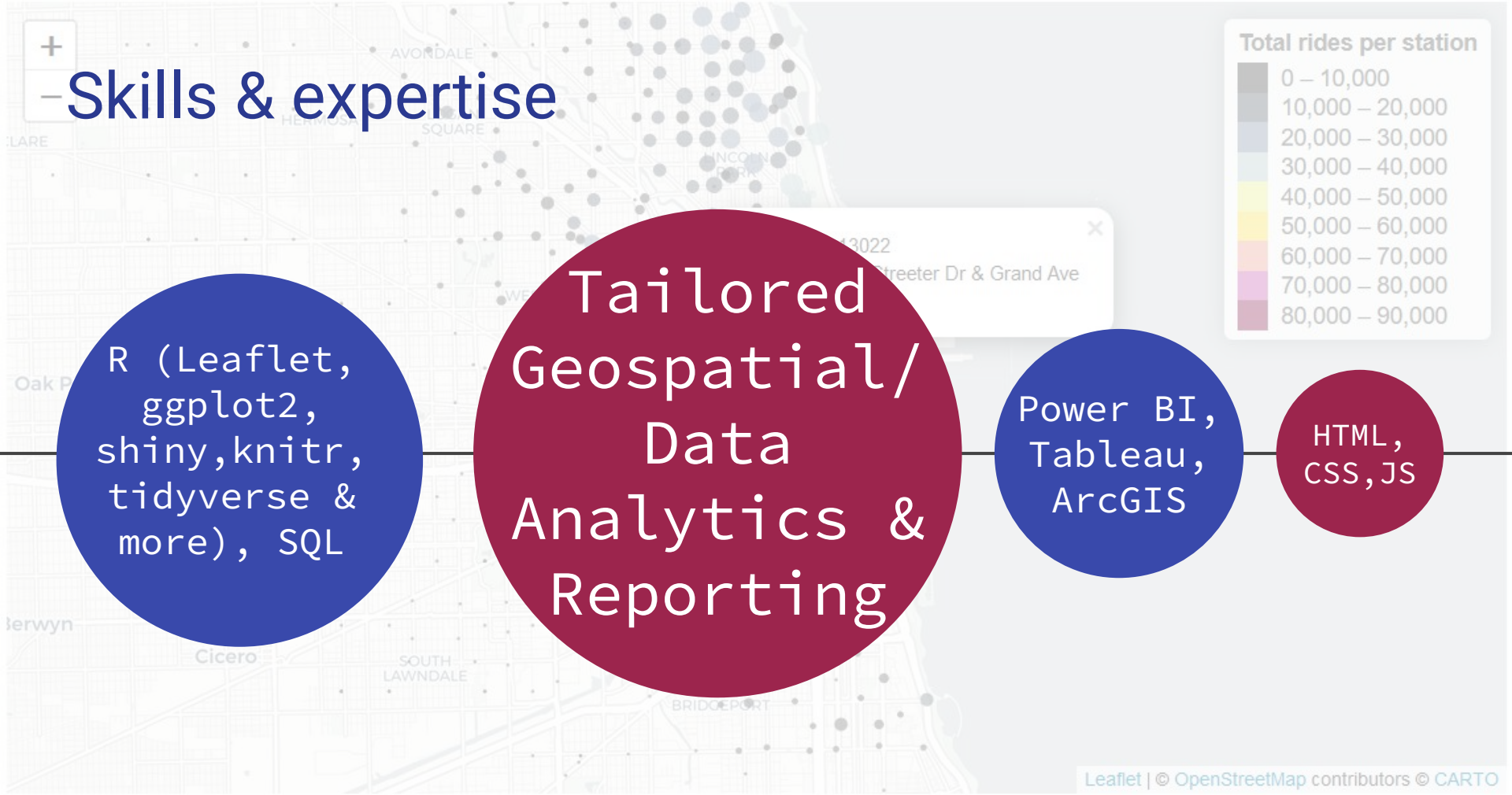
Geospatial Analytics

- **GIS Platforms:** ArcGIS Pro, QGIS, Earth Engine
- **Remote Sensing:** Satellite image processing, LiDAR, Raster analysis
- **Location Intelligence:** Spatial statistics, network & suitability analysis

BI & Visualization

- **Dashboards:** Power BI, Tableau, R Shiny, Leaflet
- **Reporting:** Rmarkdown, knitr, automated executive summary generation
- **Data Storytelling:** Communicating complex tech results in simple business terms

Skills & expertise



R (Leaflet, ggplot2, shiny,knitr, tidyverse & more), SQL

Tailored Geospatial/ Data Analytics & Reporting

Power BI, Tableau, ArcGIS

HTML, CSS,JS

Career Highlights

Professional Experience & Key Roles

JAN 2023 – PRESENT

Geospatial Analysis Consultant

UNICEF (EMOPS, New York, USA | OSE, Rome, Italy)

Subnational CCRI-DRM risk indicator development, dashboard & analytical app development, and automated risk modeling.

NOV 2019 – FEB 2022

Data Scientist / GIS & Remote Sensing Specialist

Solstice Environmental Management, Edmonton

Predictive machine learning algorithms for wetland classification, simplified visualization & Indigenous GIS capacity building.

SEP 2012 – PRESENT

Principal Consultant & Climate Hazard Modeler / Senior Partner

Jolexy Environmental Services, Edmonton

Consultancy leadership, Hydromet early warning dashboards, air quality spatial assessments & open-source R package development.

MAR 2018 – OCT 2019

GIS Analyst / Data Scientist

PLACE Research Lab, Univ. of Alberta, Edmonton

Geospatial solutions for complex public health challenges & spatial behavioral mapping.

Portfolio samples

Bike-Share Trend Analysis using R

Project Scope

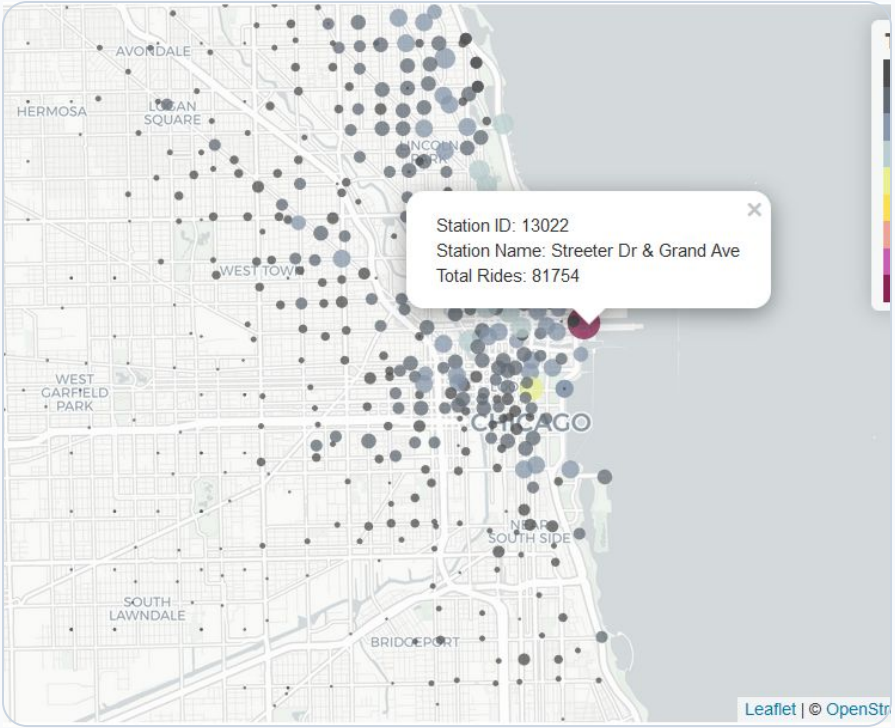
Analysis to determine bike-share trends in the Chicago area for Cyclistic (Apr 2021 – Mar 2022).

Data Source

Downloaded from Divvy Tripdata S3 under a license agreement for educational purposes.

Data Pipeline

Data download, unzipping, merging, and cleaning was entirely performed in R.



Bike-Share Trend Analysis: Descriptive R Analysis

Automated data cleaning, feature extraction, and multi-file data aggregation pipeline

01. Data Cleaning & Feature Engineering Script

```r
SA <-read_csv("Cyclistic202104_202203.csv" )
##Delete duplicate ride ids and fill in missing values using other similar rows
S1 <- SA %>%
  group_by(start_station_id,start_station_name) %>% fill(start_lat, start_lng) %>% ungroup %>%
  group_by(end_station_id,end_station_name) %>% fill(end_lat, end_lng) %>% ungroup %>%
  group_by(start_lat, start_lng)%>% fill(start_station_id,start_station_name) %>% ungroup %>%
  group_by(end_lat, end_lng) %>% fill(end_station_id,end_station_name) %>% ungroup

##distinct(ride_id, .keep_all= TRUE) %>%

#%>% group_by()
S1$RideLength <- difftime(S1$ended_at, S1$started_at,units="mins")
##Extract day of the week. 1=Sunday and 7 is Saturday
S1$day_of_the_week<-wday(S1$started_at)
S1$Hour <-hour((S1$started_at))
S1$Month<-month((S1$started_at))
S1$Wd2 <-wday(S1$started_at,label=T)
```

- **Missing Value Imputation:** Fills missing station names and coordinates using implicit directional grouping.
- **Ride Duration Calculation:** Derives exact trip duration (difftime) in minutes.
- **Temporal Feature Extraction:** Extracts day of week, hour of day, and month for downstream trend profiling.

02. File Aggregation & Dataset Compilation Script

```r
#List all files
L1 <- list.files(pattern ="20" )
##Create empty list to store column names
XLlist <-list()
##read files and extract column names
for (i in L1) {
 A1 <- read.csv(file = i, header = T)
 A2 <-  as.data.frame(colnames(A1))
XLlist[[i]] <- A2
}

XList2 <- list.cbind(XLlist)
L2 <- L1 %>%
  lapply(read_csv) %>%    # Store all files in list
  bind_rows # Combine data sets into one data set
write.csv(L2,"Cyclistic202104_202203.csv")
```

- **Directory File Search:** Dynamically scans target directory for monthly CSV trip data files.
- **Schema Inspection:** Loops through files to build uniform structure lists.
- **Row Binding & Export:** Concatenates monthly data frames into a consolidated master file (Cyclistic202104_202203.csv).

Bike-share Trend Analysis: Most Popular Station

Determining most frequent takeoff & destination stations using custom R mode calculation

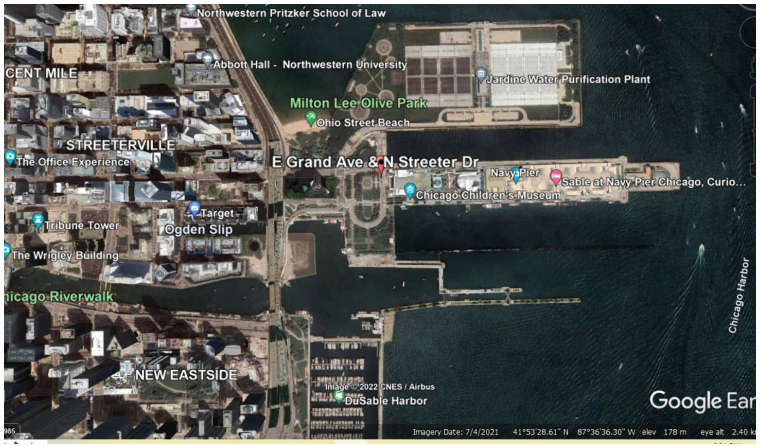
R Script Implementation (Mode Function)

```
 find the mode of the months of incidents
e = function(x){
a = table(x)
am = max(ta)
f (all(ta == tam))
 mod = NA
lse
 if(is.numeric(x))
   mod = as.numeric(names(ta)[ta == tam])
lse
 mod = names(ta)[ta == tam]
eturn(mod)


Mode <-  function(x){
a = table(x)
am = min(ta)
f (all(ta == tam))
 mod = NA
lse
 if(is.numeric(x))
   mod = as.numeric(names(ta)[ta == tam])
lse
 mod = names(ta)[ta == tam]
eturn(mod)


<- S1 %>%  filter(!start_lat >=45) %>% group_by(member_casual) %>%
ummarize(Most_popular_takeoff_Station_id=Mode(start_station_id),
         Most_popular_takeoff_Station=Mode(start_station_name),
        Most_popular_destination= Mode(end_station_name),Mosttravelled_Hour=Mode(Hour
         Weekday_most_travel= Mode(day_of_the_week),Mean_rideLength=mean(RideLength),
         Max_ridelength= as_hms(max(RideLength)),Month_Most_Travel=Mode(Month))
e.csv(S2,"Cyclistic_summary2.csv")
```

Top Station Location (Streeter Dr & Grand Ave)



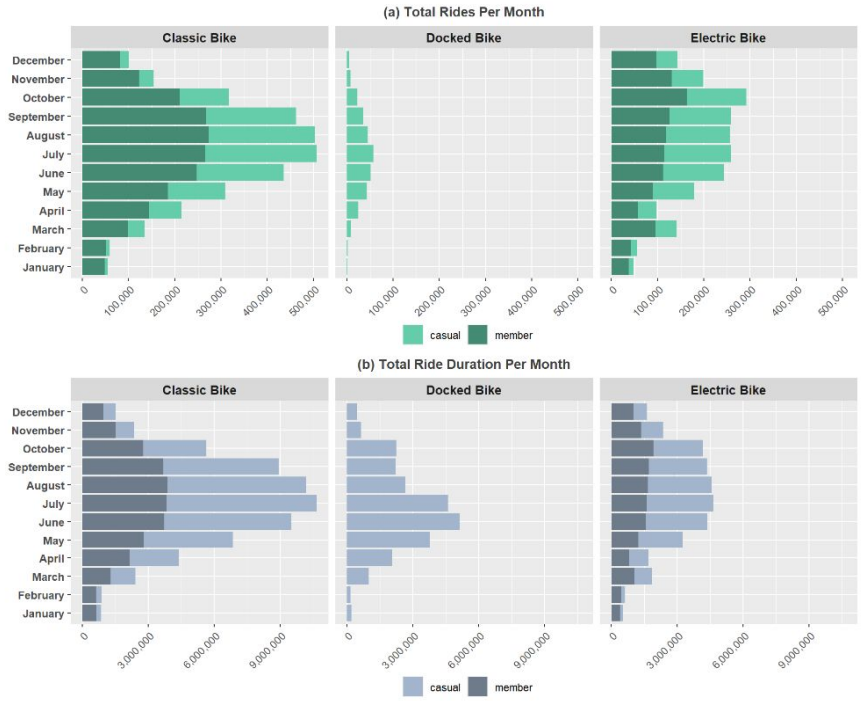
Key Observation Summary:

- **Top Station:** Streeter Dr & Grand Ave (Station ID: 13022)
- **User Insight:** Highest casual rider concentration near Navy Pier & waterfront attractions.

Bike-share Trend Analysis: Monthly Rides & Duration

Comparing monthly volume and duration trends between annual members and casual riders

Visual Analysis (Monthly Ride Metrics)



R Script Implementation (`ggplot2`)

```r
Months<- c("January","February","March","April","May","June",
           "July","August","September","October","November","December")

Bikes<-c("classic_bike"="Classic Bike","docked_bike"="Docked Bike",
         "electric_bike"="Electric Bike")
A<- ggplot(CYM, aes(fill=member_casual, y=Total_rides, x=as.factor(Month))) + theme_gray() +
  theme(legend.position = "bottom",axis.text.x = element_text(angle = 45,hjust=1,vjust=1),
        axis.text.y = element_text(face="bold", size=9),
         strip.text = element_text(face="bold", size=10),legend.margin=margin(-20, 0, 0, 0),
        plot.title = element_text(color="gray25",size=12,hjust = 0.5,face="bold"))+
  coord_flip()+ggtitle("(a) Total Rides Per Month")+
    geom_bar(position="stack", stat="identity")+
  scale_fill_manual("",values=c("aquamarine3","aquamarine4")) +
 facet_wrap(~ rideable_type,labeller = as_labeller(Bikes))+ ylab("")+
  xlab("")+scale_x_discrete(labels= Months)+ scale_y_continuous(labels = comma)


B<- ggplot(CYM, aes(fill=member_casual, y=Ride_Length, x=as.factor(Month))) +
  geom_bar(position="stack", stat="identity") +theme_gray () +
  theme(legend.position = "bottom",axis.text.x = element_text(angle = 45,hjust=1,vjust=1),
        axis.text.y = element_text(face="bold", size=9),
        strip.text = element_text(face="bold", size=10),legend.margin=margin(-20, 0, 0, 0),
        plot.title = element_text(color="gray25",size=12,hjust = 0.5,face="bold"))+
  scale_fill_manual("",values=c("lightsteelblue3","lightsteelblue4")) +
   coord_flip() +ggtitle("(b) Total Ride Duration Per Month")+
  facet_wrap(~ rideable_type,labeller = as_labeller(Bikes)) + ylab("")+
  xlab("")+scale_x_discrete(labels= Months)+ scale_y_continuous(labels = comma)

grid.arrange(A,B,ncol=1)
```

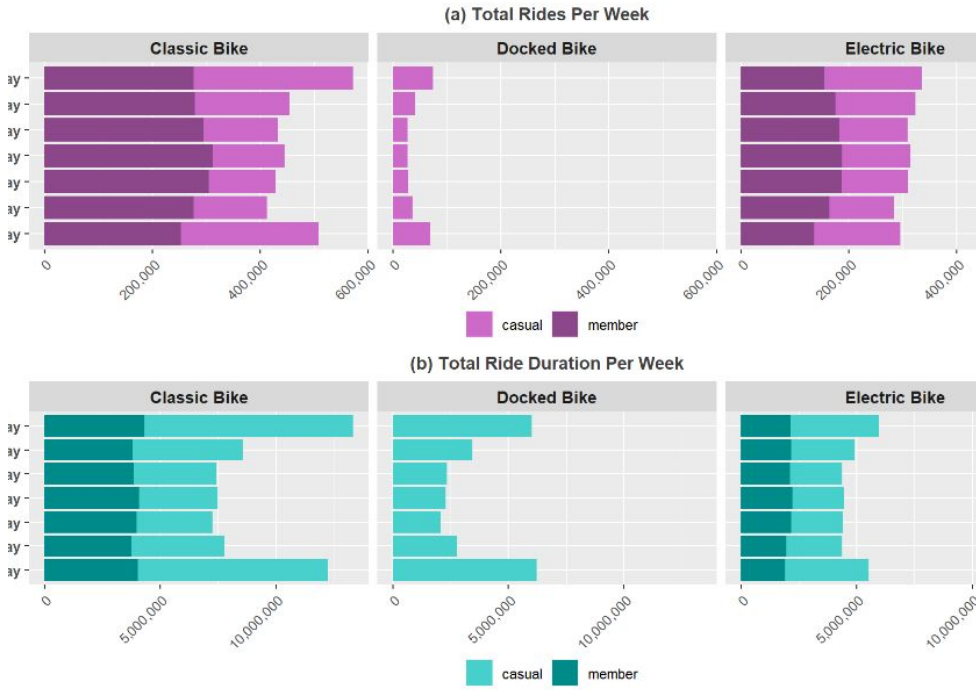
Key Observation Summary:

- **Summer Surge:** Casual rider activity and total ride duration peak drastically between June and August.
- **Consistent Utility:** Annual members demonstrate consistent usage patterns year-round, serving as primary baseline commuters.

Bike-share Trend Analysis: Weekly Bike Rides

Comparing weekly ride counts, duration patterns, and top stations between member and casual riders

Visual Analysis (Weekly Ride & Duration Metrics)



Key Observation Summary

Weekend & Tourism Surge:

More casual riders used the bike-share service in summer months and on weekends, suggesting leisure and tourism-related travel.

Station Preferences:

Casual Riders: Station 13022 (Streeter Dr & Grand Ave)

Annual Members: Kingsbury St & Kinzie St

Project Links & Resources

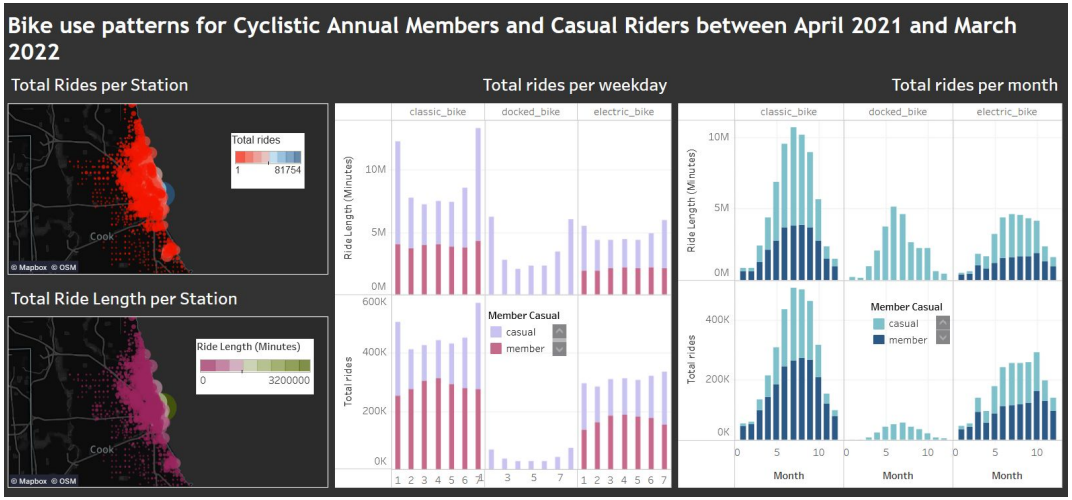
Full Report:

https://rpubs.com/aronojeghuo/Google-Cyclistic

Bike-share Trend Analysis: Tableau Visualizations

Interactive spatial and temporal data dashboard for Cyclistic ride patterns

Interactive Dashboard Overview



Dashboard Analytics Modules

- **Spatial Heatmaps:** Maps showing ride frequency and total ride length per station across Chicago.
- **Weekday Usage:** Breakdown of ride counts by day of week across classic, electric, and docked bikes.
- **Monthly Trends:** Seasonal rider patterns contrasting member vs. casual user behaviors over 12 months.

Live Tableau Dashboard

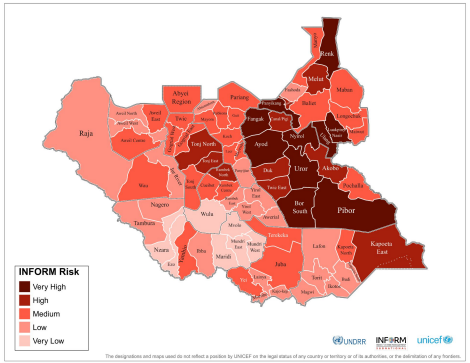
Explore the full interactive visualization online:

View Dashboard on Tableau Public

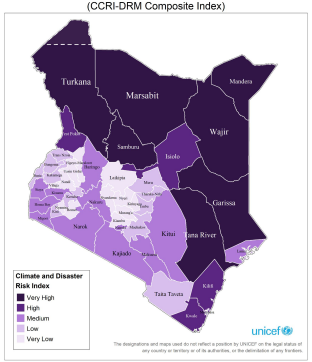
UNICEF CCRI DRM Subnational Models & Hazard Data

Subnational Risk Analysis, Climate Hazard Exposure & Disaster Risk Modeling

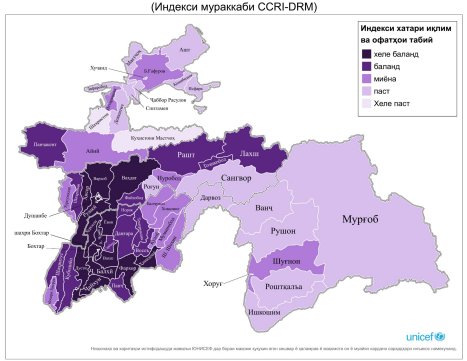
INFORM South Sudan: INFORM Risk



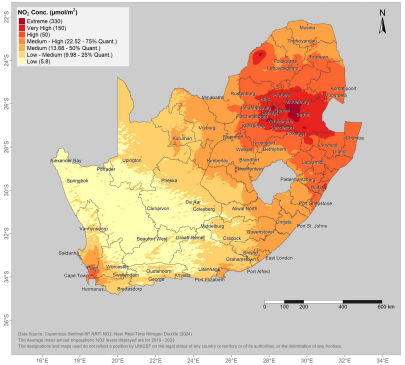
CCRI-DRM Kenya: Children's Climate and Disaster Index (CCRI-DRM Composite Index)



CCRI-DRM Тоҷикистон: Индекси хатари иқлим нисбати кӯдако - модели хатари офатҳо (Индекси мураккаби CCRI-DRM)

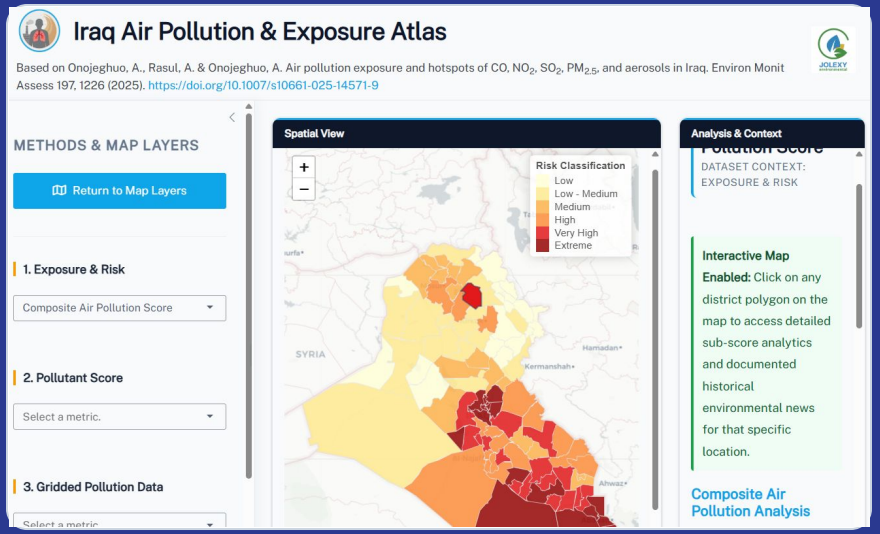


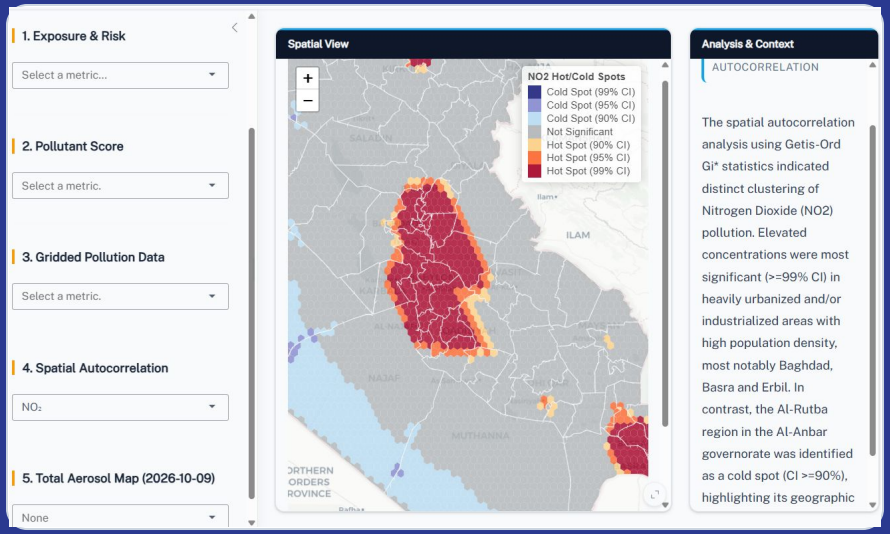
South Africa: Air Pollution - Average Annual Mean NO_2 Concentrations (2019-2023)



Iraq: Air Pollution Analysis

Interactive Environmental Data Atlas & Spatial Risk Dashboard



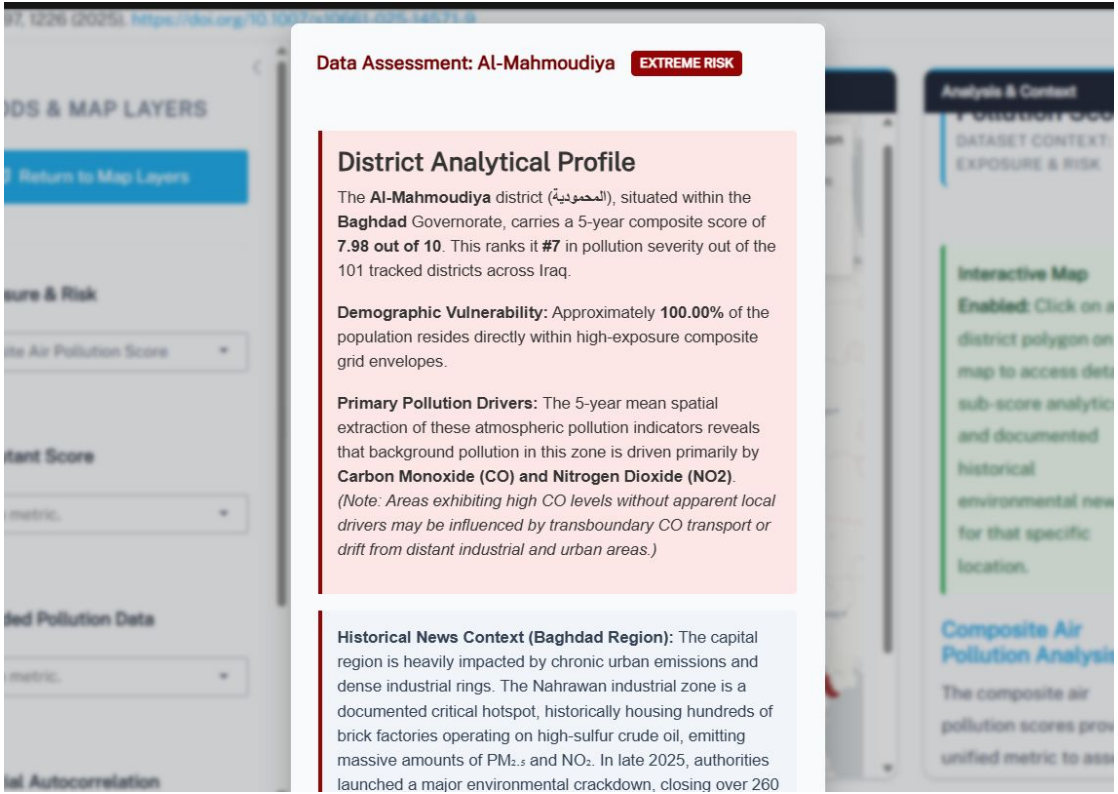


Interactive R Shiny Dashboard with NLP-based summaries: https://jolexyenviro.com/iraq-air-quality-%26-health

Publication: Based on Onojeghuo, A., Rasul, A. & Onojeghuo, A. *Air pollution exposure and hotspots of CO, NO2, SO2, PM2.5, and aerosols in Iraq.* Environ Monit Assess 197, 1226 (2025). https://doi.org/10.1007/s10661-025-14571-9

Iraq: Air Pollution Analysis

AI-Generated District & Governorate Environmental Summaries



AI CONTEXTUALIZATION

Automated District Insights

Leveraging AI-generated summaries to contextualize air pollution scores across each district and governorate.

- **District Profiling:** Rapid extraction of key atmospheric indicators (CO, NO_2, SO_2, $PM_{2.5}$).
- **Demographic Risk:** Identification of high-exposure composite grid envelopes.
- **Historical Context:** Integration of localized industrial and urban news history.

Sales Analysis using Power BI Desktop

Top Customers & Regional Purchase Value Breakdown



Full_Name	Count of OrderQuantity	City
Alvarez, Max	11	Geelong
Gao, Jon	11	St. Leonards
Huang, Eugene	11	Seaford
Miller, Edward	11	Rockhampton
Raman, Candace	11	South Melbourne
Srini, Veronica	11	Caloundra
Suarez, Melody	11	Townsville
Gutierrez, Jaime	10	Wollongong
Total	**102**	

POWER BI DESKTOP

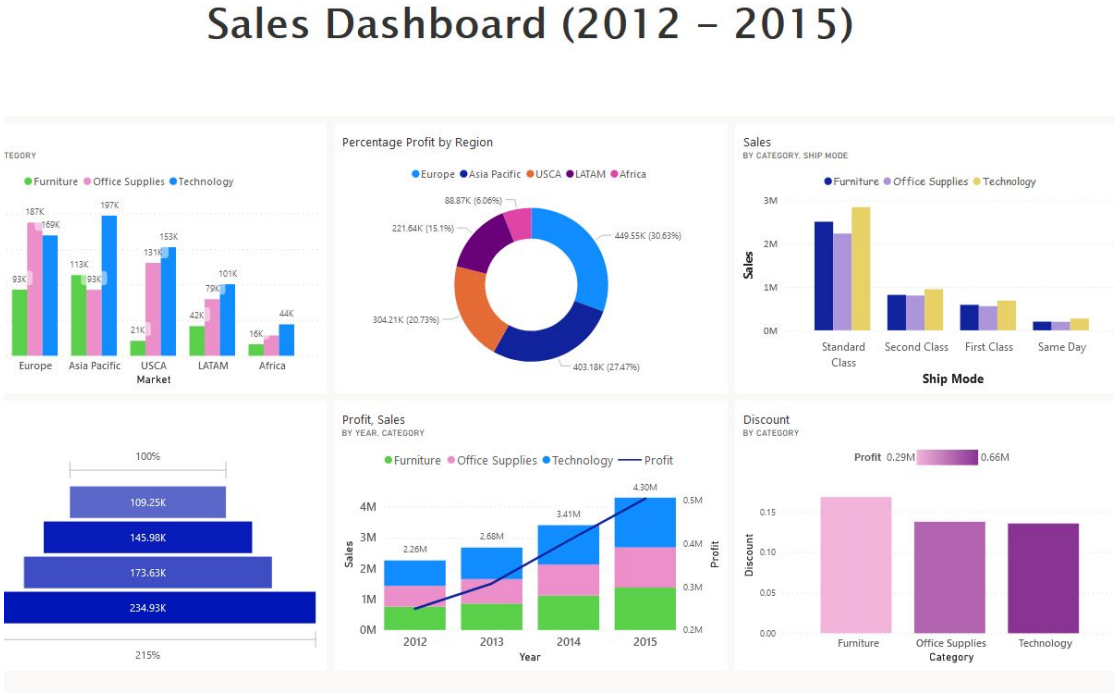
Customer & Regional Analysis

Power BI sheet showing top customers by value of total purchases per country.

- **Regional Slicer:** Interactive slider allows clients to shift between countries.
- **Customer Metrics:** Ranks top customers by order quantity and total purchase value.
- **Visual Distribution:** Integrated donut chart displays sales volume per top customer.

Sales Analysis using Power BI Service App

4-Year Profit Trend & Regional Performance Overview



POWER BI SERVICE APP

Sales & Profit Insights

Power BI App displaying interactive dashboard metrics for sales analysis, profit trends, and regional breakdowns.

- **4-Year Trend:** Tracks profit performance and trajectory from 2012 through 2015.
- **Regional Profit:** Illustrates percentage profit contribution across major global markets.
- **Category Analysis:** Evaluates sales and discount metrics across product segments.

Contact & Professional Profile

Geospatial Analysis, Climate Hazard Modeling & Environmental Data Science

GET IN TOUCH

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LOCATION & PRACTICE

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